AP CALCULUS AB

Unit 1 • Worksheet 1-6A • Answer key

Worksheet 1-6A • Average Rate of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = x^2 - 4x$ on $[1, 5]$?

(A) 2

(B) 8

(C) -2

(D) 4

Solution

Answer: (A). $f(5) = 5$, $f(1) = -3$: $\frac{8}{4} = 2$.

Question 2

NO CALCULATOR

What is the average rate of change of $g(x) = x^3$ on $[-1, 2]$?

(A) 9

(B) 3

(C) $\frac{7}{3}$

(D) 1

Solution

Answer: (B). $\frac{8-(-1)}{3} = 3$.

Question 3

NO CALCULATOR

What is the average rate of change of $h(x) = \sqrt{x}$ on $[4, 9]$?

(A) 5

(B) 1

(C) $\frac{1}{5}$

(D) $\frac{3}{9}$

Solution

Answer: (C). $\frac{3-2}{9-4} = \frac{1}{5}$.

Question 4

NO CALCULATOR

t (h)	0	3	5	9
$P(t)$ (people)	120	150	210	300

What is the average rate of change of P on $3 \leq t \leq 9$?

(A) 150 people
(B) 30 people per hour
(C) 20 people per hour
(D) 25 people per hour

Solution
Answer: (D). $\frac{300-150}{9-3} = 25$ people per hour.

Question 5

NO CALCULATOR

$C(x)$ is the cost, in FCFA, of x kilograms of rice. What are the units of $\dfrac{C(10)-C(4)}{10-4}$?

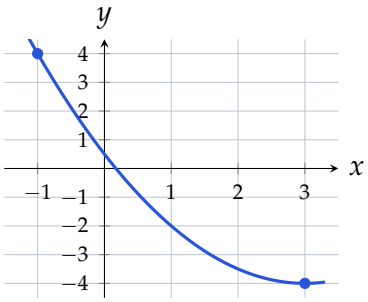
(A) FCFA
(B) FCFA per kilogram
(C) kilograms per FCFA
(D) kilograms

Solution
Answer: (B). Output units per input unit.

Question 6

NO CALCULATOR

The graph of f is shown. What is the average rate of change of f on $[-1,3]$?



(A) 2
(B) -8
(C) -2
(D) $-\frac{1}{2}$

Solution
Answer: (C). $f(-1) = 4$, $f(3) = -4$: $\frac{-8}{4} = -2$.

Question 7

NO CALCULATOR

The average rate of change of f on $[0, 4]$ is 3, and $f(0) = 5$. What is $f(4)$?

(A) 17
(B) 12
(C) 8
(D) 7

Solution
Answer: (A). $\frac{f(4)-5}{4} = 3$, so $f(4) = 17$.

Question 8

NO CALCULATOR

On a graph, the average rate of change on $[a, b]$ is the slope of

(A) the tangent line at a
(B) the x-axis
(C) the tangent line at b
(D) the secant line through $(a, f(a))$ and $(b, f(b))$

Solution
Answer: (D). Slope between the two points.

Question 9 (Free response)

NO CALCULATOR

A tank is drained.

t (min)	0	10	20	30	40
$V(t)$ (L)	500	420	360	320	300

(a) Find the average rate of change of V on $[0, 40]$. Give units and interpret it.
(b) Find the average rate of change of V on $[20, 40]$.
(c) Find the average rates on $[0, 10]$, $[10, 20]$, $[20, 30]$, $[30, 40]$. Is the water leaving faster or slower over time?

Solution
(a) $\frac{300-500}{40} = -5$ L/min: on average the tank loses 5 liters per minute.
(b) $\frac{300-360}{20} = -3$ L/min.
(c) -8, -6, -4, -2 L/min: the rates increase toward 0, so the water leaves more and more slowly.